Predictive Thermal Battery Management System for Li-ion Battery Packs

Abstract

An internal short inside one cell is often named as the cause of lithium-ion battery accidents. In its most common form it starts small: it slowly drains its own cell and makes only a fraction of a watt of heat. A normal battery management system (BMS) cannot see this stage, because it acts only when a voltage, current or temperature crosses a fixed limit. In this project I am building a four-cell (4S1P) NCR18650GA battery pack with two layers of protection: an independent hardware BMS, and an ESP32 supervisor that predicts. Five times a second, the ESP32 runs a physical thermal model of every cell with a Kalman filter, and estimates what no sensor can measure: the core temperature, and the abnormal heat, in watts, that the cell's own current cannot explain. It forecasts each cell's temperature 1 and 5 minutes ahead and checks every forecast against the real reading, and at rest it estimates each cell's self-discharge from its voltage. All of this evidence becomes one risk level (Normal, Watch, Warning or Trip), which can open separate charge and load relays and keep them latched.

I benchmarked the design against six commercial products, from a ₹239 hardware BMS board and protection chips from TI, ADI and NXP to the Orion BMS and the Li-ion Tamer gas sensor. All of them handle temperature with fixed limits, and the only early warning among them, the gas sensor (US$596), reacts only after a cell has vented. Among seven published methods, the longest warnings come from methods trained on cells or a vehicle taken into thermal runaway, and only one of them acts on the battery.

The Delta X over the base paper, Ojo et al. (IEEE TIE, 2021), which flags faults with a trained neural network, is as follows. My design needs no training data: 11 physical constants, fitted once from normal use, replace about 21,000 trained weights. The base paper's 0.69 °C threshold within its 4.5 s window catches heating only above about 9 °C/min, while my design is built to warn at 0.5 °C/min (about 0.4 W), 18 times slower, and its 5-minute check catches 0.2 W. It adds electrical evidence of a short, a leak from 20 mA; it needs 10–30 s of persistence and agreement between thermal and electrical evidence before an alarm; it uses about 2,000 times less arithmetic per estimate on a ₹639 ESP32 instead of a Jetson TX2 or Raspberry Pi; and it disconnects the pack instead of only raising a flag.

Semester 1 builds and calibrates the electronics and the first predictive firmware with no lithium cells (Gates 0–2). Semester 2 builds the pack and tests every target at Gates 9–11, including a safely emulated soft short. Until those tests, all figures for my design are targets, not results.

Ms. Muskan Sulathana (23WH1A0208)

Signature

Dr. M. Rupesh

Associate Professor

EEE Department